

Problem Definition

In this project, we are not supposed to use real data. There is no CLIP embedding to the word “Nupzuki” so image generated by giving prompt “Nupzuki” to the SDXL doesn’t match with *real* “Nupzuki” character. In this scenario, our work is to **make the image of “Nupzuki” character without explicit real image**. Furthermore, our work is to make various concepted “Nupzuki” images **only with SDXL base model and pretrained LoRAs**.

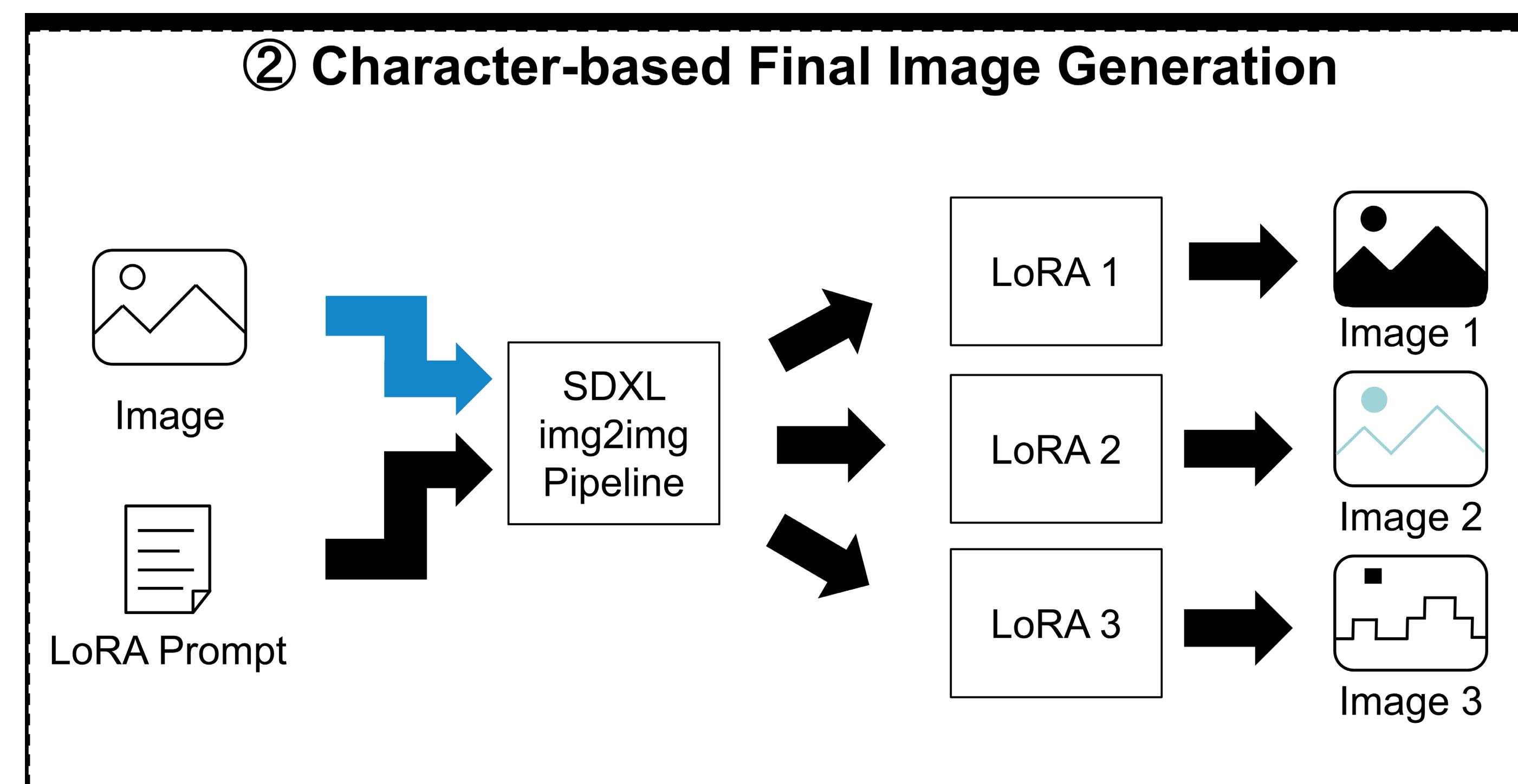
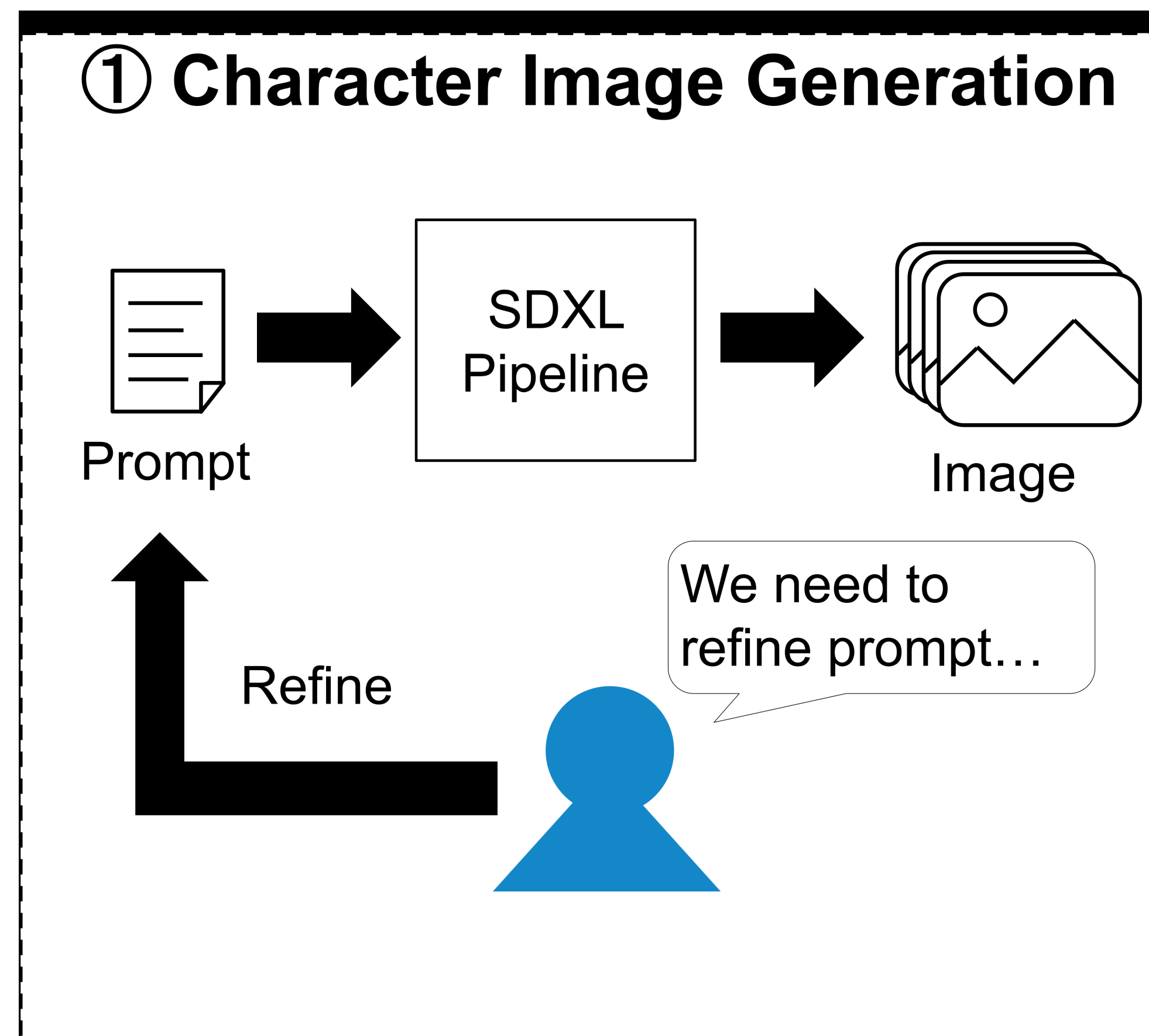
Key Ideas

- Human Feature Extraction
- Prompt Engineering
- Image, text Conditioned Diffusion Generation
- Image Variation by Utilizing LoRA

References

[1] huggingface Diffusers Stable Diffusion XL, https://huggingface.co/docs/diffusers/en/api/pipelines/stable_diffusion/stable_diffusion_xl
[2] SDXL, Podell, D., et al (2024). SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis.

Method



Results



Our pipeline **successfully generate** various “**Nupzuki**”s!